

Standard electrode potentials (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to interpret standard electrode potential data in terms of ease of oxidation/reduction.

In your studies so far, you have been studying the reduction and oxidation of different species in a qualitative manner. You previously studied the ease with which some metals are oxidised compared to others and organised this information into a reactivity series (see [section R3.2.1–2 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/)). Though this allows you to compare the reactivity of one metal to another, it is important to be able to quantify how much more reactive one element is than another in a redox reaction.

Take for example a common non-rechargeable battery (see [section R3.2.3–4 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ease-of-oxidation-and-reduction-id-46488/)). These types of batteries are composed of a variety of different metals including cadmium, lead, zinc, manganese, nickel, silver, mercury and lithium. Why are these metals chosen to be used in batteries? How can you quantify the amount of electricity that could be generated when certain metals are combined in a voltaic cell?

In this section you will study the concept of standard electrode potentials and use these to determine the standard cell potential for a voltaic reaction such as those that occur in many primary cells.

Using the hydrogen half-cell potential (SHE)

In any situation where a measurement is recorded you must have an appropriate measuring device with an appropriate unit of measurement assigned to it.

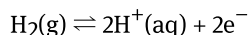
For example, temperature can be measured using various techniques and units. One scale commonly used is Celsius or centigrade which is based around the temperature at which water solidifies and evaporates. These two reference points were arbitrarily set at 0.0 °C and 100.0 °C. This notion of setting arbitrary standard values as reference points is one that is essential to measurements in electrochemistry.

In order to quantify whether one element will more readily undergo reduction or oxidation compared to another element, we need a standard value around which to base our measurements. This comes in the form of the standard hydrogen electrode (SHE).

The standard hydrogen electrode is made of a platinum electrode submerged in a solution of 1.00 mol dm⁻³ H⁺(aq) ions ([H⁺(aq)] = 1.00 mol dm⁻³). Hydrogen gas at 100 kPa is then bubbled over the platinum electrode and a redox reaction occurs. Platinum is used as it is inert and will not preferentially undergo reduction over hydrogen gas.

In the reaction, hydrogen ions are reduced to form hydrogen gas and hydrogen gas is oxidised to produce hydrogen ions in solution. When we combine the two half-reactions we get a net reaction that appears to be reversible. This is one of the reasons that hydrogen was chosen as the standard upon which to base all other values.

The hydrogen half-cell



is assigned an arbitrary standard electrode potential ($E^\ominus$) of zero volts (0V) by convention.

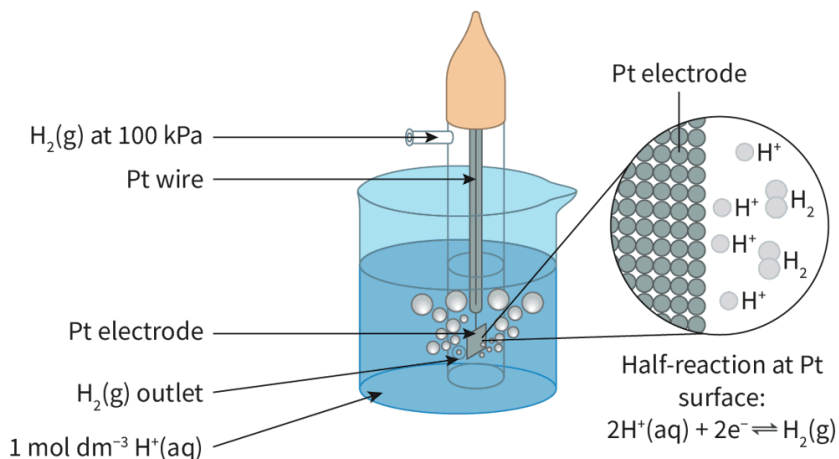


Figure 1. A standard hydrogen electrode.

 More information for figure 1

The diagram depicts a standard hydrogen electrode. A platinum electrode is submerged in an aqueous solution containing $1 \text{ mol dm}^{-3} \text{H}^+$ ions. Hydrogen gas (H_2) at 100 kPa is bubbled over the electrode. The diagram shows the gas entering through an outlet and a platinum wire connecting to the electrode. On the right, an enlarged section illustrates the half-reaction at the platinum surface: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{H}_2(\text{g})$. The components are labeled, including Pt electrode and Pt wire.

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Theory of Knowledge

A platinum electrode is a key requirement of the standard hydrogen electrode. Platinum is an incredibly hard and expensive metal with a high melting point (1775°C). Ancient societies in present-day Colombia and Ecuador worked with platinum to make jewellery, masks and vessels, yet platinum was largely unknown in Europe until the mid-sixteenth century. These ancient metallurgists were capable of making objects and purifying this unreactive metal without a furnace capable of reaching the melting point of platinum. To what extent does creativity and persistence play in the pursuit of knowledge?

So now that you are aware of this arbitrarily assigned standard value around which all standard electrode potentials ($E^\ominus$) are derived, how can you use this information?

As you have seen previously electrochemical reactions can be described in terms of their half-reactions (see [section R3.2.1–2](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/>)). These half-reactions are each associated standard electrode potentials ($E^\ominus$), measured in volts (V), at standard ambient temperature and pressure (SATP). It is not possible to obtain the standard electrode potential of a single half-cell reaction as voltages are measurements of potential difference between two locations, the anode and the cathode. Because of this issue, the SHE is used in conjunction with a metal electrode to determine the relative reducing power of different half-cells compared to hydrogen. Depending on the reducing power of the metal electrode, the SHE is then forced to act as either a cathode or an anode when connected to the other half-cell.

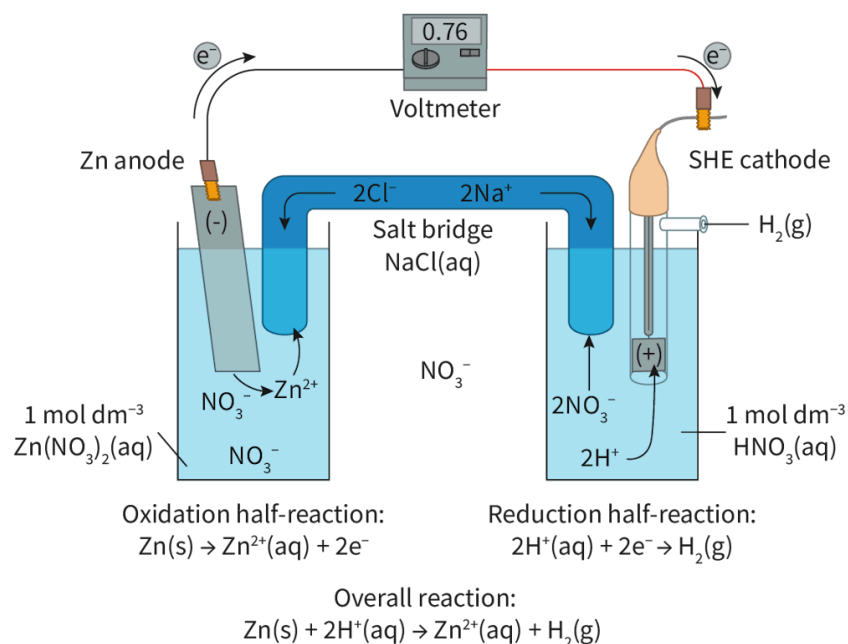


Figure 2. Determining the standard electrode potential for zinc using the SHE.

 More information for figure 2

The image is a diagram of an electrochemical cell setup to measure the standard electrode potential for zinc using the Standard Hydrogen Electrode (SHE) as the reference. On the left, a container holds the zinc anode, represented as Zn(s) , immersed in a zinc nitrate solution ($1 \text{ mol dm}^{-3} \text{ Zn(NO}_3)_2(\text{aq})$). At the anode, zinc undergoes oxidation according to the half-reaction: $\text{Zn(s)} \rightarrow \text{Zn}^{2+}(\text{aq}) + 2\text{e}^-$. On the right is another container housing the SHE cathode where the reduction half-reaction occurs: $2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow \text{H}_2(\text{g})$.

A voltmeter connects the two half-cells indicating the electric potential, with a salt bridge containing NaCl solution (2Cl^- and 2Na^+) linking them to allow for the movement of ions and complete the circuit. Arrows indicate electron flow from the zinc anode to the SHE cathode, and the setup is designed to measure the potential difference when zinc is oxidized, making the zinc half-cell act as the anode while the SHE half-cell acts as the cathode.

The overall electrochemical reaction can be summarized as: $\text{Zn(s)} + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2(\text{g})$.

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This allows us to determine the standard electrode potentials for various species and rank these species against one another in terms of how easily they are oxidised or reduced. As you will notice in **Table 1**, these reactions are listed as reduction half-reactions of metal ions forming a reduced metal product.

Table 1. Examples of various standard electrode potentials.

Half-reaction	$E^\ominus$
$\text{Ni}^{2+} + 2\text{e}^- \rightarrow \text{Ni}$	-0.23 V
$\text{Cu}^+ + \text{e}^- \rightarrow \text{Cu}$	$+0.52\text{ V}$
$\frac{1}{2}\text{Cl}_2 + \text{e}^- \rightarrow \text{Cl}^-$	$+1.36\text{ V}$

These values, when ranked in order from most negative to most positive, produce a reactivity series, which can be found in the DP chemistry data booklet. The values are always given as the reagent species being reduced within the reduction half-reaction. The number of moles of electrons and reactants and products need not be equal to zero.

Species that have a more negative standard electrode potential will be more easily oxidised. This also means that they will be stronger reducing agents, and that they will be placed higher in the reactivity series.

Species with a more positive standard electrode potential will be more easily reduced. They will be stronger oxidising agents and will also be ranked lower in the reactivity series.

Lithium has the most negative standard electrode potential of all metals, at -3.03 volts, compared to gold which has one of the most positive values at $+1.50$ volts.

Use **Interactive 1** to rank the metals in order of how easily they are oxidised.

Easily oxidised



Easily reduced

☒ Check

Interactive 1. Ranking metals according to ease of oxidation.

Making connections

You previously studied the concept of ionisation energy (see [section R3.1.2—3](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/conjugate-acids-bases-id-46053/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/conjugate-acids-bases-id-46053/>) and variable oxidation states of transition metals (see [section R3.1.9—10](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ion-dissociation-constants-of-weak-acids-and-bases-hl-id-46058/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ion-dissociation-constants-of-weak-acids-and-bases-hl-id-46058/>)). Recall that transition metals have variable oxidation states due to the fact that their ionisation energies are close in magnitude. It is interesting to note that there is some correlation between the sign of the standard electrode potential of a certain species and its ionisation energy or energies.

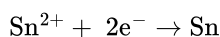
Note that lithium has the lowest first ionisation energy of all metal species and also has the most negative standard electrode potential!

Standard electrode potentials, like changes in enthalpy and equilibrium constants are examples of state functions. This means that if you reverse a particular half-cell reaction then the standard electrode potential for that reaction will be reversed in sign, but its magnitude will stay the same. Though individual half-reactions are reversible, the direction of the overall net reaction for an electrochemical cell depends on what specific combination of electrodes and solutions are being used to construct the cell. The setup of the cell determines the direction in which the reaction will proceed, and from this setup one can deduce which species are undergoing reduction and oxidation.

Study skills

Standard electrode potentials can sometimes seem confusing. It is important to remember that these potentials are reduction potentials for reduction half- reactions that produce a product with a more negative oxidation state.

When tin (Sn) is said to have a standard electrode potential value of -0.14 V this is for the half-reaction:

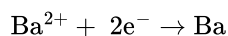


not for tin itself being reduced.

Worked example 1

Barium has an $E^{\ominus} = -2.90\text{ V}$.

Give the half-cell reaction that has this value of standard electrode potential.



where:

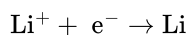
$$E^{\ominus} = -2.90\text{ V}$$

The most commonly occurring cation of barium is Ba^{2+} . Therefore, the half-reaction that forms barium from those barium ions and two electrons has a standard electrode potential of -2.90 V .

Worked example 2

Lithium metal easily undergoes oxidation when compared to all other metals.

Give the standard electrode half-cell reaction for lithium and predict the sign for its standard electrode potential.



where:

$$E^{\ominus} = \text{a large negative value}$$

Standard electrode potentials are always listed as reduction half-reactions, and so the half-reaction for lithium involves a lithium cation gaining one electron to form a lithium atom.

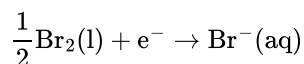
Since lithium is more easily oxidised than all other metals, you can deduce that it must have an value that is large and negative, placing it at the very top of the reactivity series.

Worked example 3

Bromine can be reduced to form bromide ions in solution.

Give the half-reaction for the reduction of bromine to produce one mole of bromide ions.

Predict the sign for the value for the half-reaction.



where:

$E^\ominus$ = a large positive value

In this case the bromide ion will have a -1 oxidation state and be aqueous on the products side of the equation. The ion is formed from bromine gas, which is a diatomic molecule in liquid form at SATP, and has an oxidation state of 0. To balance the reaction to produce one mole of bromide ions, there must only be a half mole of liquid bromine as the reactant gains one mole of electrons.

This reaction will have a large positive standard electrode potential.

Bromine is easily reduced to form bromide ions (recall that bromine has a high electron affinity), which means it will be found towards the bottom of the reactivity series and have a positive value.

Now that you have a better understanding of standard electrode potentials, let's apply these new concepts.

Activity

- **IB learner profile attribute:** Inquirers
- **Approaches to learning:** Thinking skills — Designing procedures and models, Experimenting with new strategies for learning
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Individual/pair activity

Using the simulator, you (and a partner if working in pairs) will set up an electrochemical reaction using two half cells. Once you have constructed the appropriate voltaic cells you will record the necessary qualitative and quantitative data so as to answer a series of questions related to standard electrode potentials.

Your task

1. Access the simulator via the following link:
<https://teachchemistry.org/classroom-resources/voltaic-cells> 
(<https://teachchemistry.org/classroom-resources/voltaic-cells>)
2. Under the heading Galvanic/Voltaic Cells click on Start →
3. Create a data table in Excel or a digital notebook, consisting of a column with the heading 'Metal Electrodes' and containing four different metal electrodes (Ag, Cu, Zn and Mg), a second column with the heading 'Voltage' and a third column with the heading 'Molecular Half-reaction'.
4. Using the simulator, construct a voltaic cell consisting of a hydrogen electrode in HCl in the left beaker and a silver electrode in $\text{AgNO}_3(\text{aq})$ in the right beaker.

5. Start the simulation by clicking the button on the voltmeter from off to on.
6. Record any quantitative data (including measurement uncertainties).
7. Click on the 'See Molecular Scale' button and then record the half-reaction that is taking place at the surface of the silver electrode.
8. Repeat the experiment three more times using the Cu, Zn and Mg electrodes and keeping the hydrogen electrode constant in the left-hand beaker.

Use the data recorded during the activity to answer these questions.

1. Explain how it is possible to determine the standard electrode potentials ($E^\ominus$) for each metal in this experiment, using the data you (and your partner) collected.
2. Describe the percent uncertainty of your results and describe how the percent uncertainty might change if you were to conduct this experiment in the laboratory and not using a simulator.
3. List the half-cell reactions occurring at the metal electrode in order from most easily oxidised to most easily reduced.
4. Hypothesise as to what effect reversing the beakers for the hydrogen and metal electrodes would have on the standard electrode potential's magnitude and/or sign.
5. Compare and contrast your results with other members of your class.